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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-47

Filter Inductor Design

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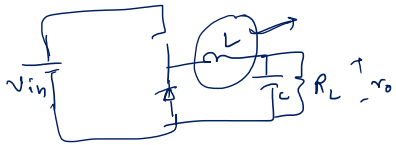


Recap

- Transformer design procedure

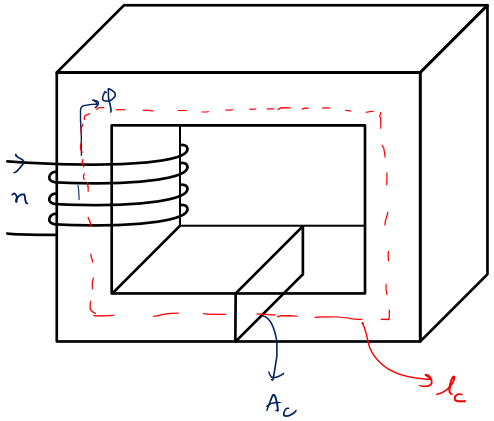
select the core → $\Delta B_{,opt}$ → number of turns in primary winding → numbers of turns in other winding → evaluate the wiresizes which is needed to actually wound the winding ($\rho_{j,opt}$)

$P_{total} \leq P_{total,allow}$



Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Inductor



① Ampere's law

$\Rightarrow H_c l_c = n i$

($H_c \rightarrow$ the magnetic field intensity inside the core)

② The flux density, $B = \mu_c H_c$

$\Rightarrow$ flux, $\phi = B \cdot A_c$

$\Rightarrow \phi = A_c \cdot \mu_c H_c$

$\Rightarrow \phi = \dfrac{\mu_c A_c n i}{l_c} \longrightarrow (1)$

($\mu_c = \mu_r \mu_0$; $\mu_0 = 4\pi \times 10^{-7}$; $\mu_r \rightarrow$ relative permeability)

$\Rightarrow \mu_c \gg \mu_0$

($n \rightarrow$ no. of turns; $i \rightarrow$ current excitation)

③ flux linkage, $\lambda = n\phi$

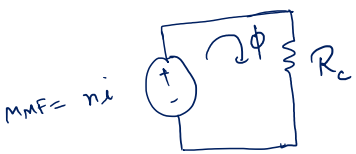
$\Rightarrow \lambda = \dfrac{\mu_c A_c n^2 i}{l_c}$

④ inductance, $L = \lambda / i \Rightarrow L = \dfrac{\mu_c A_c n^2}{l_c} \longrightarrow (3)$

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Inductor

Magnetic circuit model



⇒ electrical equivalence ($\phi \to i$, $R_c \to R$, MMF → v)

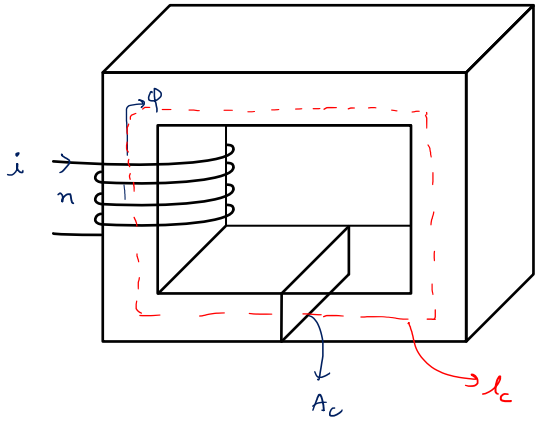
⇒ $\phi = \frac{MMF}{R_c} = \frac{ni}{R_c}$ → (2)

⇒ from (1) & (2)

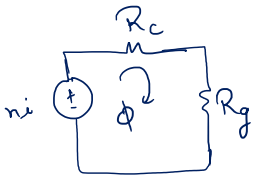
⇒ $R_c = \frac{l_c}{\mu_c A_c}$ → (4)

⇒ from (3) & (4)

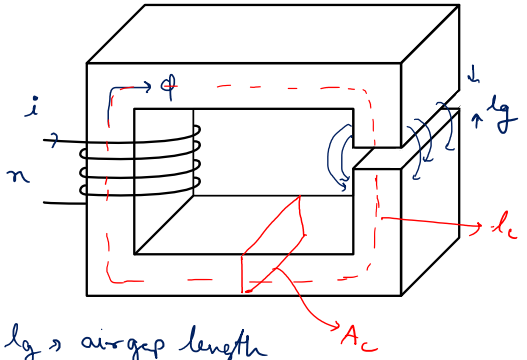
⇒ $L = \frac{n^2}{R_c}$ ⇒ if we know the equivalent reluctance offered to flux path using this reluctance, we can calculate inductance of any complex magnetic structure



Inductor



$$\Rightarrow \phi = \frac{ni}{R_c + R_g} \Rightarrow \phi = \frac{ni}{\frac{l_c}{\mu_c A_c} + \frac{l_g}{\mu_0 A_g}} \rightarrow (5)$$



$l_g \rightarrow$ air gap length

$A_g \rightarrow$ area of cross section offered by air gap

$$\Rightarrow R_{eq} = R_c + R_g$$

$$\Rightarrow L = \frac{n^2}{R_{eq}} = \frac{n^2}{\frac{l_c}{\mu_c A_c} + \frac{l_g}{\mu_0 A_g}}$$

$\Rightarrow A_c \approx A_g$

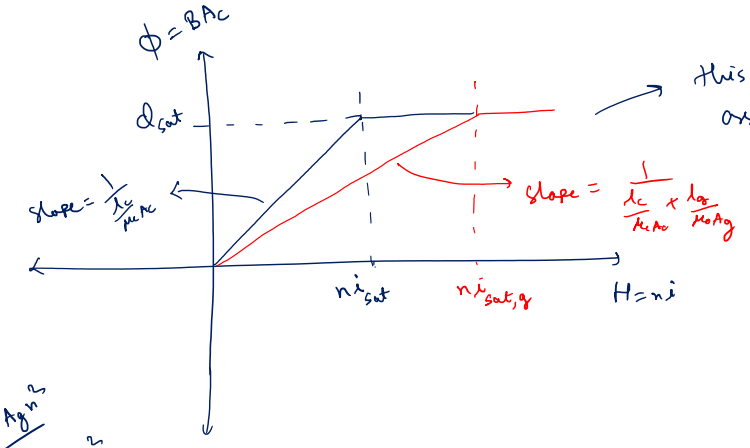
$\Rightarrow A_c \neq A_g$

$$R_c = \frac{l_c}{\mu_c A_c} \; ; R_g = \frac{l_g}{\mu_0 A_c}$$

$\mu_c \gg \mu_0 \Rightarrow R_c \ll R_g$

$$\Rightarrow L = \frac{n^2}{\frac{l_g}{\mu_0 A_g}} \Rightarrow \boxed{L = \frac{\mu_0 A_g n^2}{l_g}}$$

Inductor



this curve is drawn with assumption that the non-linearities are absent

$L_g = \frac{\mu_0 A_g n^2}{l_g}$

$L_c/l_g = \frac{\mu_c A_c n^2}{l_c}$

$i_{sat,g} >> i_{sat}$

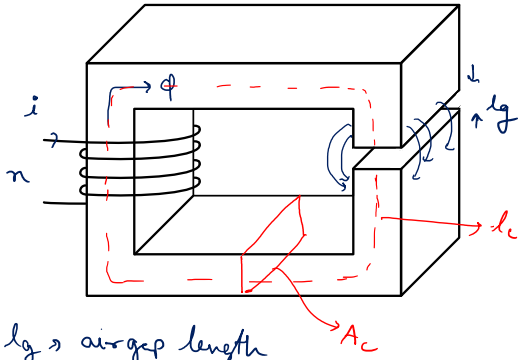
$\mu_c >> \mu_0$

$\Rightarrow \frac{l_c}{\mu_c} << \frac{l_g}{\mu_g}$

$\Rightarrow$ slope with gap $= \frac{\mu_0 A_g}{l_g}$

$\Rightarrow$ Air-gap is introduced

- $\rightarrow$ Inductance value goes down
- $\rightarrow$ the current at which the core saturates increase
- $\rightarrow$ the inductance is independent on the core dimensions

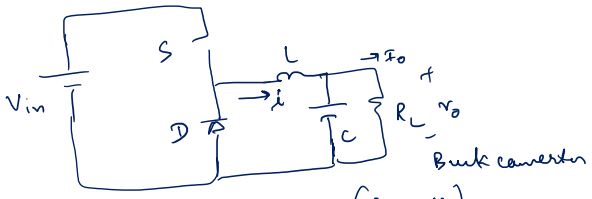


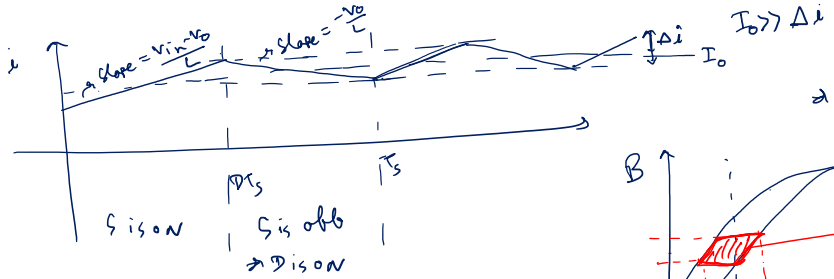
$l_g \rightarrow$ air-gap length

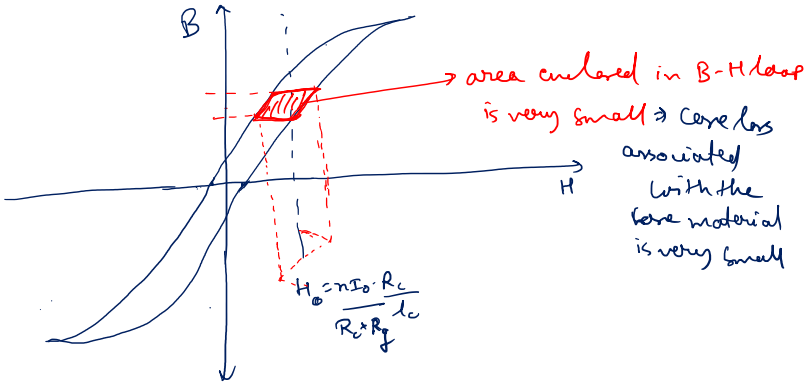
$A_g \rightarrow$ area of cross section offered by air gap

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

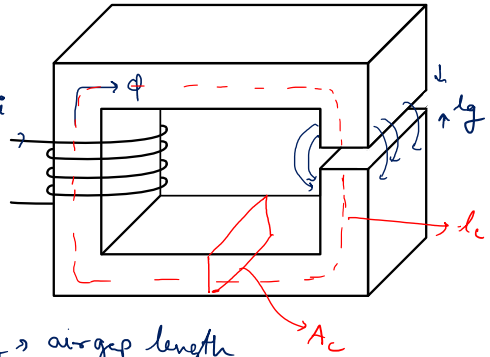
Filter Inductor











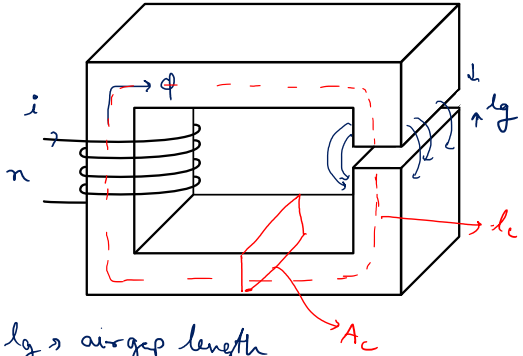
$l_g \Rightarrow$ air-gap length

$A_g \Rightarrow$ area of cross section offered by air gap

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Filter Inductor

- Since the Area enclosed in the B-H loop is small, the Core loss in the inductor is negligible
- Since the ripple component in inductor current is small, the winding AC resistance is negligible.
  - (skin effect or proximity effect)
- the flux density is choosen based on saturation value (B_{sat})
- Air gap is employed to avoid saturation



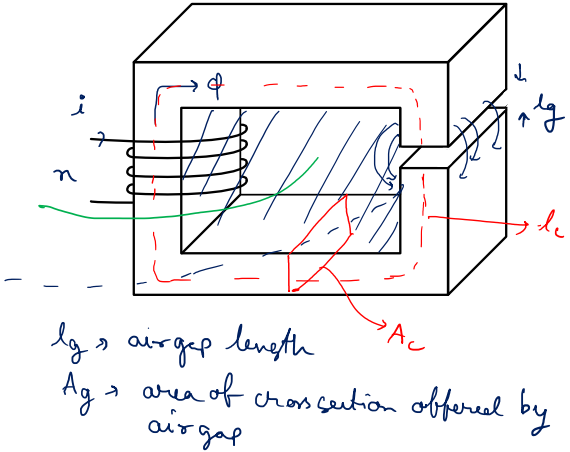
lg → air gap length

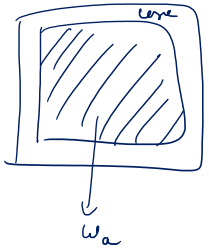
Ag → area of cross section offered by air gap

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Filter Inductor Design

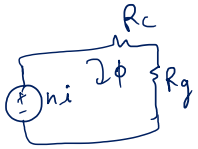
- Design Outcome
  - Core size (core cross-sectional area, window area, mean length per turn, magnetic path length)
  - Number of turns
  - Wire size
  - Air-gap length
- Specification
  - Required inductance: L
  - Allowed power loss: $P = I_{rms}^2 R$
  - Maximum current: I_{max}





Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Saturation



$\Rightarrow ni = \phi(R_c + R_g)$

as $R_c << R_g$

$\Rightarrow ni = \phi R_g$

$\Rightarrow nI_{max} = \phi_{max} R_g$ $\quad \left(R_g = \frac{l_g}{\mu_0 A_g}\right)$

$\Rightarrow nI_{max} = B_{max} A_c R_g$

$\Rightarrow nI_{max} = B_{max} A_c \cdot \frac{l_g}{\mu_0 A_g}$

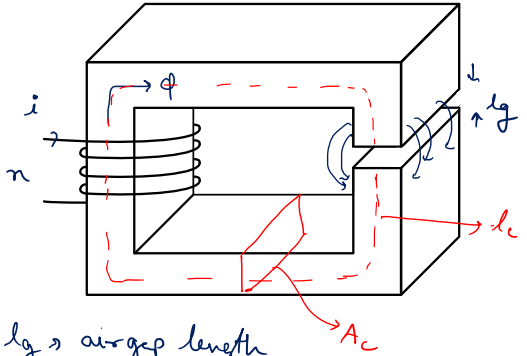
$(A_c \simeq A_g)$ → (ie the bringing of flux is very less)

$\Rightarrow nI_{max} = \frac{B_{max} l_g}{\mu_0}$

$\Rightarrow B_{max} < B_{sat}$ (maximum flux density must be less than the B_{sat})

B_{sat} for the ferrite core $0.25\,T$

$\simeq B_{max} \simeq 0.2$ to 0.22



l_g → air gap length

A_g → area of cross section offered by air gap

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Inductance value

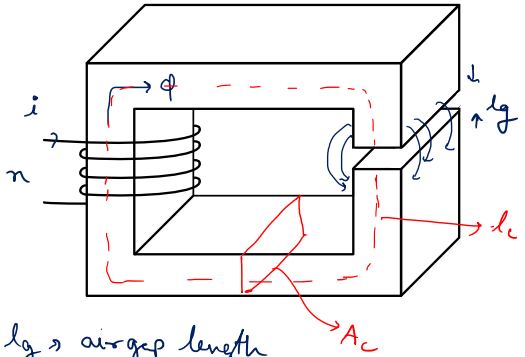
- The inductance is due to the air-gap

$$L = \frac{\mu_0 A_c n^2}{l_g}$$

⇒ the fringing effect is neglected ⇒ $A_c = A_g$

⇒ $L = \frac{\mu_0 A_c n^2}{l_g}$

$\mu_0 = 4\pi \times 10^{-7}$



l_g → air-gap length

A_g → area of cross section offered by air gap

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

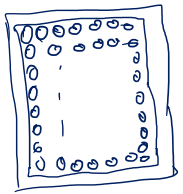
Winding Area

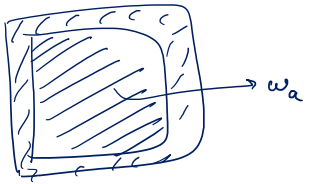
- If the winding has n turns, then the area of copper conductor in the window is nA_W $A_w \rightarrow$ area of cross section of the wire used for winding
- Area available for the winding is $k_u w_a$
- Thus, the third design constraint is

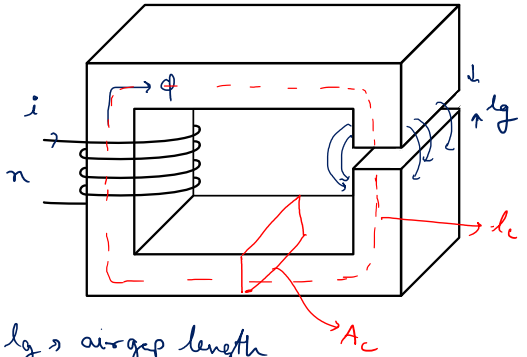
$$k_u w_a \geq nA_W$$

↳ winding fill factor









$l_g \rightarrow$ air gap length

$A_g \rightarrow$ area of cross section offered by air gap

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Winding Resistance

- The resistance of the winding is

$$R = \rho \frac{n(MLT)}{A_w}$$

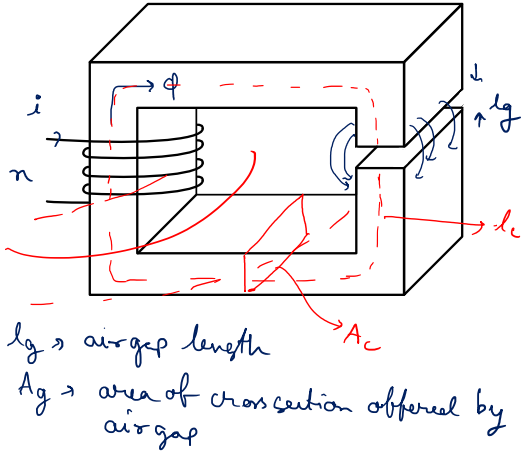
n → no. of turns

AC effects are neglected

A_w

$R = \rho \frac{l}{A_w}$

$\Rightarrow R = \rho \frac{n(MLT)}{A_w}$



Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Summary

(1) $nI_{max} = \frac{B_{max} l_g}{\mu_0} \rightarrow (1)$ $(B_{max} < B_{sat})$

(2) $L = \frac{\mu_0 A_c n^2}{l_g} \rightarrow (2)$ (inductance due to air-gap)

(3) $K_u W_a \geq n A_w \rightarrow (3)$

(4) $R = \rho \frac{n(MLT)}{A_w} \rightarrow (4)$

Rearrange (1), (2), (3), (4)

$$\frac{A_c^2 W_a}{MLT} \geq \frac{\rho L^2 I_{max}^2}{B_{max}^2 R K_u} \rightarrow \text{specified value} \rightarrow (5)$$

Core dimensions

→ I_{max}, B_{max}, μ_0 ($4\pi \times 10^{-7}$), L, K_u, R, ρ (material of wire) are obtain from specification, core material and circuit operation.

→ n, l_c, MLT, W_a, A_c and A_w which need to be known

Inductor Design Procedure

Step-1

Select the core → l_c, A_c, w_a, MLT

$$\frac{A_c^2 w_a}{MLT} \geq \frac{L^2 I_{max}^2}{B_{max}^2 R K_u}$$ (this condition has to be met) → (1)

EE5
EE10
EE25
⋮
EE45

⇒ we need to select the smallest core whose dimensions (l_c, A_c, w_a, MLT) satisfies the condition in eq. (1)

→ l_c, A_c, w_a, MLT

Step-2

$$LI = n\phi$$

$$\Rightarrow LI_{max} = n \cdot B_{max} A_c$$

$$\Rightarrow n = \frac{LI_{max}}{B_{max} A_c}$$

⇒ the no. of turns to be used, $n = \frac{LI_{max}}{B_{max} A_c}$

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Inductor Design Procedure

Step-3

the air gap, $l_g = \frac{\mu_0 A_c n^2}{L}$

Sometimes, the manufacture provides inbuild airgap

then they provide A_L (mH/1000turns)

$$\Rightarrow A_L = \frac{B_{max}^2 A_c^3}{L I_{max}^2} \quad (\text{mH}/1000\text{turns})$$

Core size
↓
number of turns
↓
airgap length
↓
A_w (wire size)

Step-4

Evaluating the wire cross sectional area

$$\Rightarrow A_w \leq \frac{K_u W_a}{n}$$ (Assume no AC effects)

using this A_w value we can select appropiate wire having required AWG or SWG values → wire gauges

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Design Check

- The winding resistance can be calculated

$$R = \frac{\rho n (MLT)}{A_w} (\Omega)$$

- Check if it is less than the defined copper loss

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Thank You